PrivateCall — How to make a call

No account, installation, terminal, or programming knowledge is needed. Both participants simply open the HTML file.

Before you start

- Both of you need the file **PrivateCall.html** and an internet connection. Open it in Google Chrome (the browser PrivateCall was tested with).
- You also need another way to send each other a message, such as a chat app or e-mail.
- To find a connection, PrivateCall contacts public helper servers run by Google and Cloudflare. Some networks also need a relay server (see *Troubleshooting*).

Person A — starts the call

1. Open PrivateCall.html and click **“Start a call”**.
2. Choose **“Video call”** or **“Voice call”**. When the browser asks, allow the microphone (and the camera).
3. Wait a moment. On some networks this takes up to 30 seconds.
4. Under **“Send this invitation”**, click **“Copy”**. Send the whole message to Person B in any app. (Or click **“Save as file”** and send the file.)
5. Keep the window open. When the reply arrives, copy all of it. Under **“Add their reply”**, click **“Paste”**. You can also paste into the box, or click **“Open file”**. The call starts by itself.

Person B — joins the call

1. Open PrivateCall.html and click **“Join a call”**.
2. Copy the whole invitation you received. Under **“Add the invitation”**, click **“Paste”**. You can also paste into the box, or click **“Open file”**.
3. Click **“Join with camera”** or **“Join with microphone only”** (for a voice call: **“Join the voice call”**). When the browser asks, allow the microphone (and the camera).
4. Under **“Send this reply back”**, click **“Copy”** and send the whole reply to Person A.
5. Wait. The screen says: *“Waiting for them to add your reply. The call starts automatically.”*

Copy the whole message

Each message starts with a sentence, for example *“Join my private call: open PrivateCall, choose “Join a call”, and paste this whole message.”* Then comes the line `PrivateCall.invite.v1:` (or `PrivateCall.reply.v1:`), lines of letters and numbers, and finally `:end`. **Copy and send everything, from the first sentence to `:end`.** Do not change anything. An invitation can be used for one call only.

Check the security code (both of you)

When the call connects, both screens show the same 6-digit code, for example **482 917**. Until you click **“Codes match”**, the other person can’t see you, and can hear you only while your microphone is turned on in this code panel.

- **Person A** clicks **“Turn on microphone”** and says the **first** three digits: *“four eight two”*.
- **Person B** clicks **“Turn on microphone”** and says the **last** three digits: *“nine one seven”*.
- Each of you compares what you hear with your own screen. **Only if everything matches**, click **“Codes match”**. Each of you clicks it on your own screen; your camera and microphone are sent normally only after you do.

If the codes are different, click **“Codes don’t match”**. The call ends. Do not continue. Start a new call, and send the invitation through a different app if you can.

A stronger check: a skilled attacker could imitate a voice. For more certainty, compare the code as text in a different app you trust: click **“Copy code”** and send it there. Both codes must be the same.

During the call

“Microphone” turn your microphone off or on (or press M)
“Camera” turn your camera off or on (or press V)
“Leave” end the call
“Details” information about the connection

If you see **“Click to turn on sound”**, click it to hear the other person.

Troubleshooting

Permission denied. The message says PrivateCall wasn’t allowed to use your microphone or camera. Click the icon in the browser’s address bar, allow access, then try again.

No sound. Click **“Click to turn on sound”** if you see it. Check your speaker or headset volume. Before the other person clicks **“Codes match”**, you normally hear them only while their microphone is turned on in the code panel.

Invitation or reply not accepted. Copy the whole message again, from the first sentence to `:end`, or send it as a file (**“Save as file”** and **“Open file”**). The invitation goes into **“Join a call”**; the reply goes back to the computer that started the call. A used invitation does not work again: ask for a new one.

“Couldn’t connect”. Start again and exchange a new invitation and reply quickly, within a few minutes. Try a different network for one of you, such as a phone hotspot or home Wi-Fi.

When a relay may be needed. If calls keep failing, especially on office, school, mobile or other restricted networks, you may need a relay server. Someone you trust who runs one can give you a relay code that starts with `PCRELAY1:`. Open **“Connection settings”**, paste the code, and click **“Use this relay”**. Only one of you needs it.